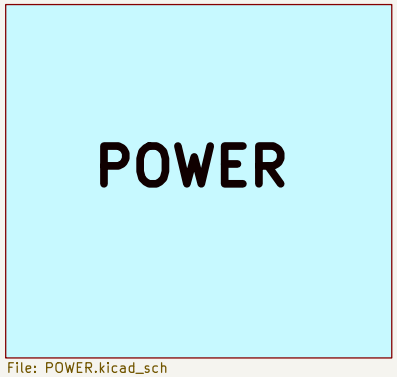
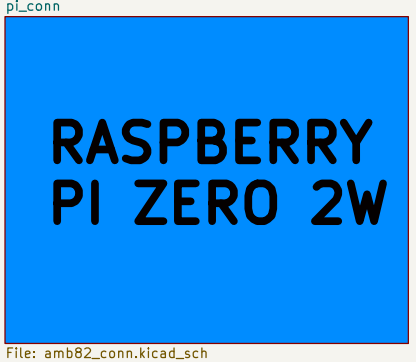


File: POWER.kicad_sch

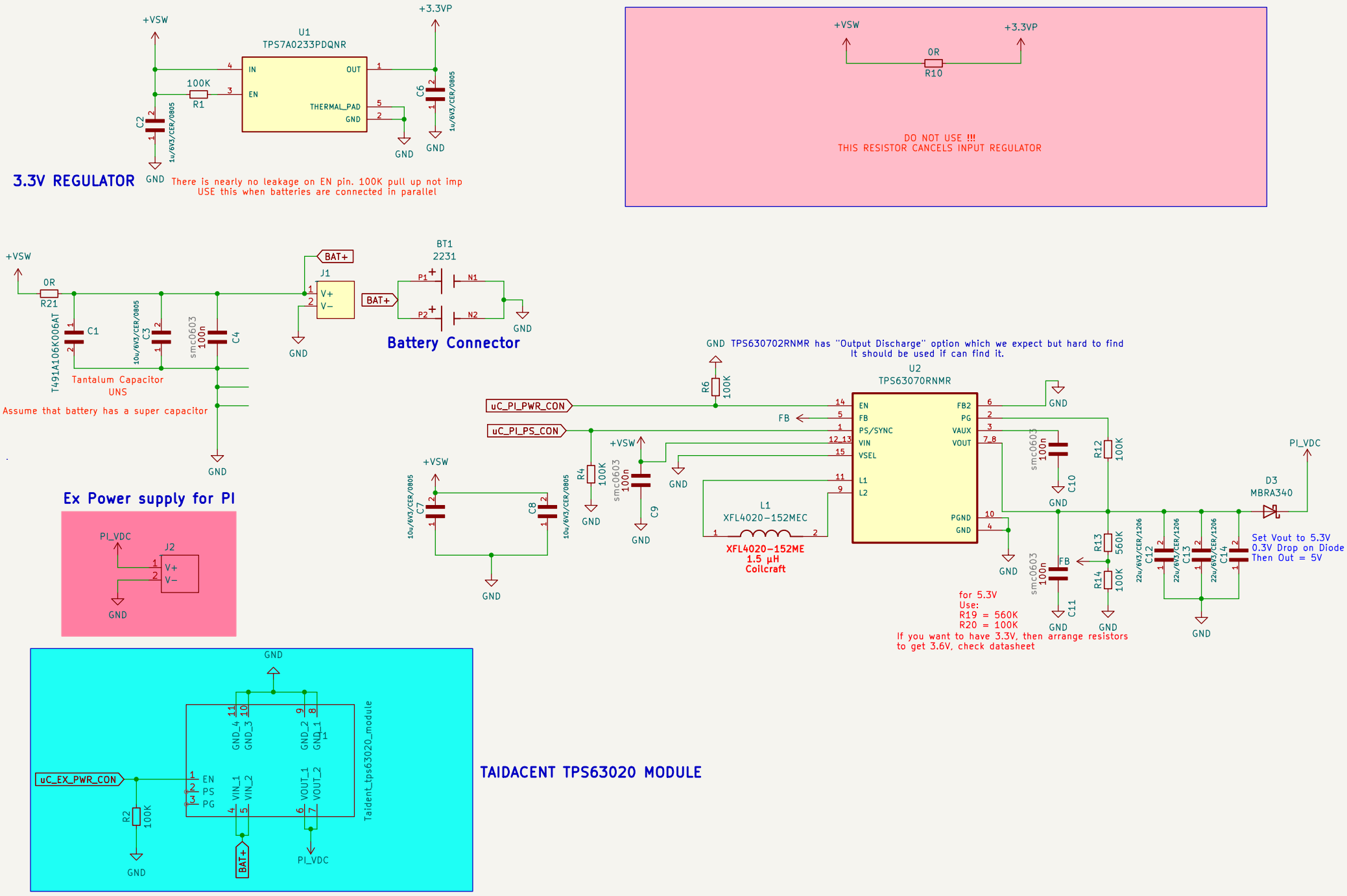


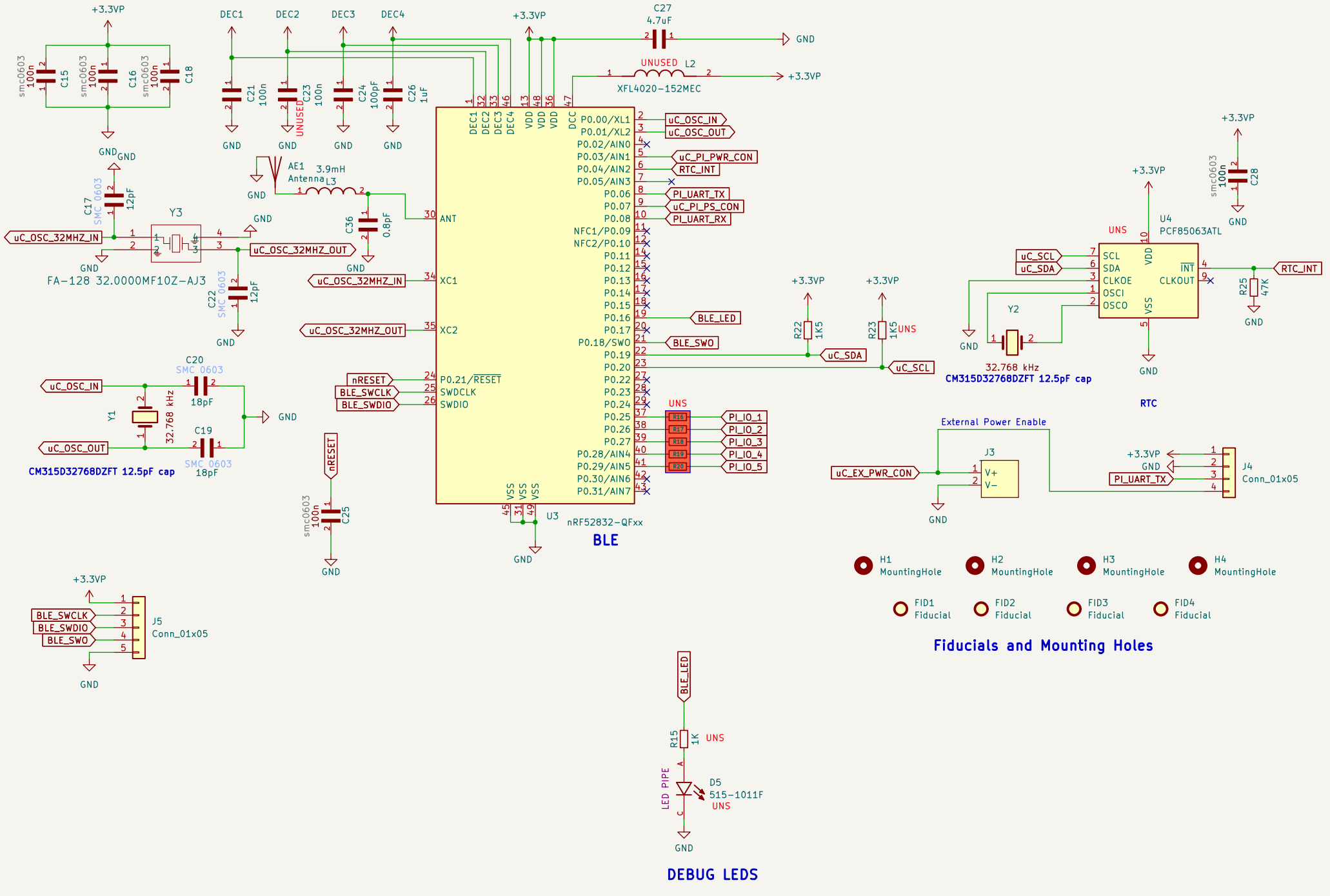
File: CPU.kicad_sch

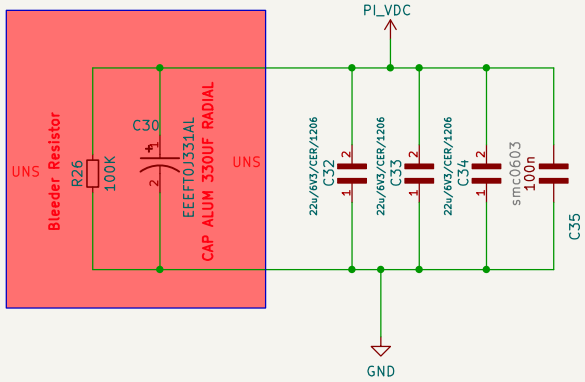
pi_conn

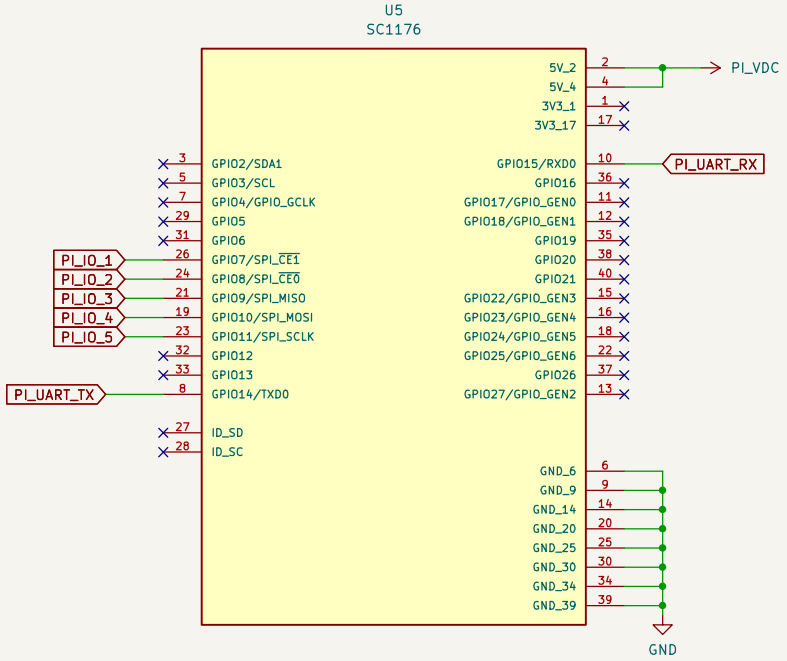


File: amb82_conn.kicad_sch


+VSW
U1
TPS7A0233PDQNR
IN
EN
OUT
THERMAL_PAD
GND
100K
R1
C2
1u/6V3/CER/0805
+3.3VP
C6
1u/6V3/CER/0805
GND
3.3V REGULATOR
There is nearly no leakage on EN pin. 100K pull up not imp
USE this when batteries are connected in parallel
+VSW
0R
R10
+3.3VP
DO NOT USE !!!
THIS RESISTOR CANCELS INPUT REGULATOR
+VSW
0R
R21
C1
T491A106K006AT
Tantalum Capacitor
UNS
C3
10u/6V3/CER/0805
C4
smc0603
100n
Assume that battery has a super capacitor
BAT+
J1
V+
V−
BT1
2231
P1
P2
N1
N2
Battery Connector
GND TPS630702RNMR has "Output Discharge" option which we expect but hard to find
It should be used if can find it.
U2
TPS63070RNMR
R6
100K
uC_PI_PWR_CON
uC_PI_PS_CON
FB
EN
PS/SYNC
VIN
VSEL
L1
L2
PGND
FB2
PG
VAUX
VOUT
L1
XFL4020-152MEC
XFL4020-152ME
1.5 µH
Coilcraft
R4
100K
C7
C8
C9
C10
C11
R12
100K
R13
560K
R14
100K
C12
C13
C14
22u/6V3/CER/1206
D3
MBRA340
PI_VDC
Set Vout to 5.3V
0.3V Drop on Diode
Then Out = 5V
for 5.3V
Use:
R19 = 560K
R20 = 100K
If you want to have 3.3V, then arrange resistors
to get 3.6V, check datasheet
Ex Power supply for PI
PI_VDC
J2
V+
V−
GND
TAIDACENT TPS63020 MODULE
uC_EX_PWR_CON
R2
100K
EN
PS
PG
VIN_1
VIN_2
VOUT_1
VOUT_2
GND_1
GND_2
GND_3
GND_4
BAT+
PI_VDC
Taident_tps63020_module


+3.3VP
C15
C16
C18
smc0603
100n
DEC1
DEC2
DEC3
DEC4
C21
C23
C24
100pF
C26
1uF
UNUSED
GND
C27
4.7uF
UNUSED L2
XFL4020-152MEC
AE1
Antenna
3.9mH
L3
C17
SMC 0603
12pF
Y3
uC_OSC_32MHZ_IN
uC_OSC_32MHZ_OUT
FA-128 32.0000MF10Z-AJ3
C22
C36
0.8pF
ANT
XC1
XC2
C20
18pF
Y1
32.768 kHz
C19
CM315D32768DZFT 12.5pF cap
nRESET
BLE_SWCLK
BLE_SWDIO
P0.21/RESET
SWDCLK
SWDIO
C25
U3
nRF52832-QFxx
BLE
P0.00/XL1
P0.01/XL2
P0.02/AIN0
P0.03/AIN1
P0.04/AIN2
P0.05/AIN3
P0.06
P0.07
P0.08
NFC1/P0.09
NFC2/P0.10
P0.11
P0.12
P0.13
P0.14
P0.15
P0.16
P0.17
P0.18/SWO
P0.19
P0.20
P0.22
P0.23
P0.24
P0.25
P0.26
P0.27
P0.28/AIN4
P0.29/AIN5
P0.30/AIN6
P0.31/AIN7
uC_PI_PWR_CON
RTC_INT
PI_UART_TX
uC_PI_PS_CON
PI_UART_RX
BLE_LED
BLE_SWO
uC_SDA
uC_SCL
PI_IO_1
PI_IO_2
PI_IO_3
PI_IO_4
PI_IO_5
R22
1K5
R23
UNS
U4
PCF85063ATL
RTC
Y2
32.768 kHz
CM315D32768DZFT 12.5pF cap
R25
47K
External Power Enable
J3
V+
V-
uC_EX_PWR_CON
J4
Conn_01x05
PI_UART_TX
J5
Conn_01x05
BLE_SWCLK
BLE_SWDIO
BLE_SWO
H1
MountingHole
H2
MountingHole
H3
MountingHole
H4
MountingHole
FID1
Fiducial
FID2
Fiducial
FID3
Fiducial
FID4
Fiducial
Fiducials and Mounting Holes
R15
1K
LED PIPE
D5
515-1011F
DEBUG LEDS





Use a paralel resistance to charge and discharge capacitors safely
!!
Observe transient with an oscilloscope
and set R and C to get a proper time constant.
MAY BE EEUFS0J472B and a Bleeder Resistor
!!

If we use TPS630702RNMR then we will not need that resistor as it has
Quick Output Discharge (QOD) ability